

Práctica: Configuración y Administración de Servidor Apache

1. Verificación de Servicios

Enunciado: Demostrar que los servicios Apache, PHP y MySQL están instalados y en funcionamiento, mostrando sus versiones y estado actual.

Pasos y Comprobaciones:

1. Apache:

- Ejecuta: `apache2 -v`
- Ejecuta: `systemctl status apache2`

```
migueldaw@miguelillo:~$ apache2 -v
Server version: Apache/2.4.58 (Ubuntu)
Server built:   2025-08-11T11:10:09
migueldaw@miguelillo:~$
```

```
apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
Active: active (running) since Wed 2025-11-26 20:40:08 UTC; 6 days ago
Docs: https://httpd.apache.org/docs/2.4/
Process: 4921 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
Process: 5318 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
Main PID: 4924 (apache2)
Tasks: 7 (limit: 2265)
Memory: 18.6M (peak: 22.1M)
CPU: 489ms
CGroup: /system.slice/apache2.service
├─4924 /usr/sbin/apache2 -k start
├─5324 /usr/sbin/apache2 -k start
├─5325 /usr/sbin/apache2 -k start
├─5326 /usr/sbin/apache2 -k start
├─5327 /usr/sbin/apache2 -k start
├─5328 /usr/sbin/apache2 -k start
└─6103 /usr/sbin/apache2 -k start

Nov 26 20:40:08 miguelillo systemd[1]: Starting apache2.service - The Apache HTTP Server:
Nov 26 20:40:08 miguelillo apachectl[4923]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.0.1. Set the 'ServerName' directive globally to suppress this message
Nov 26 20:40:08 miguelillo systemd[1]: Started apache2.service - The Apache HTTP Server
Nov 03 19:01:34 miguelillo systemd[1]: Reloading apache2.service - The Apache HTTP Server
```

2. PHP:

- Ejecuta: `php -v`

```
miguel daw@miguelillo:~$ php -v
PHP 8.3.6 (cli) (built: Jul 14 2025 18:30:55) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.3.6, Copyright (c) Zend Technologies
with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies
```

3. MySQL:

- Ejecuta: `mysql -V`
- Ejecuta: `systemctl status mysql`

```
miguel daw@miguelillo:~$ mysql -V
mysql Ver 8.0.44-0ubuntu0.24.04.1 for Linux on x86_64 ((Ubuntu))
```

```
miguel daw@miguelillo:~$ systemctl status mysql
mysql.service - MySQL Community Server
  Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: en>
  Active: active (running) since Wed 2025-11-26 20:39:26 UTC; 6 days ago
  Process: 4477 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=e>
  Main PID: 4486 (mysqld)
  Status: "Server is operational"
  Tasks: 38 (limit: 2265)
  Memory: 364.6M (peak: 380.8M)
  CPU: 29.487s
  CGroup: /system.slice/mysql.service
          └─4486 /usr/sbin/mysqld

Nov 26 20:39:25 miguelillo systemd[1]: Starting mysql.service - MySQL Community>
Nov 26 20:39:26 miguelillo systemd[1]: Started mysql.service - MySQL Community>
lines 1-14/14 (END)
```

2. Cambio del Directorio Raíz (DocumentRoot)

Enunciado: Cambiar la carpeta raíz por defecto (`/var/www/html`) a una nueva llamada DAW dentro de esa ruta, desplegando allí la web de ejemplo.

Configuración:

1. Creamos la carpeta: `sudo mkdir /var/www/html/DAW`

```
miguel daw@miguelillo:/var/www/html$ sudo mkdir /var/www/html/DAW
```

2. Copiamos/Movemos tu web de ejemplo dentro de esa carpeta.
3. Editamos la configuración del sitio: `sudo nano /etc/apache2/sites-available/000-default.conf`

Modificamos la línea DocumentRoot:

Apache

DocumentRoot /var/www/html/DAW

```
<VirtualHost *:80>
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html/DAW

    # Available loglevels: trace8, ..., tracel, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

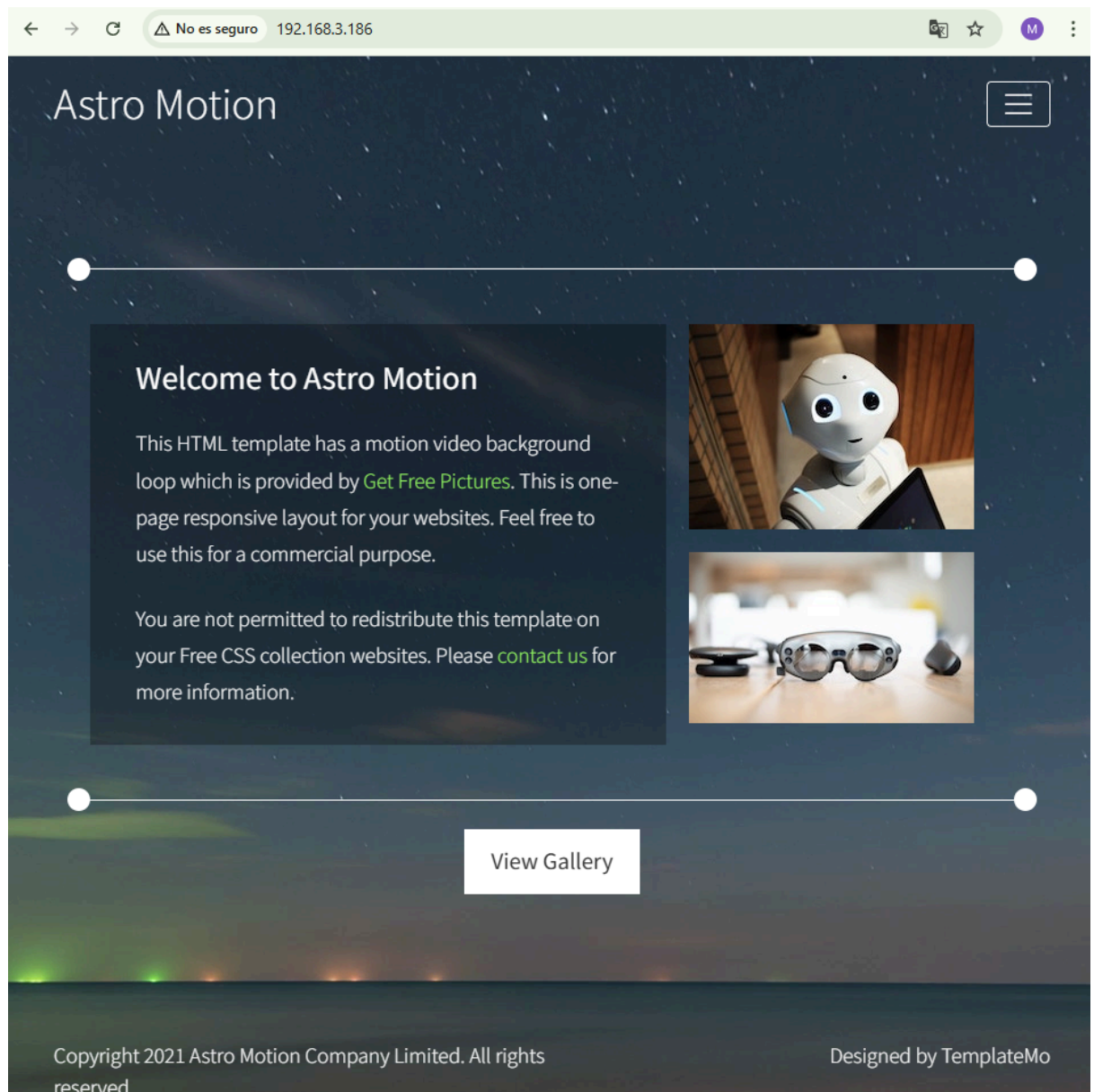
    ErrorLog ${APACHE_LOG_DIR}/error.log
```

4. Reiniciamos Apache: `sudo systemctl restart apache2`

```
miguel daw@miguelillo:/var/www/html$ sudo systemctl restart apache2
```

Prueba de funcionamiento:

- Abre el navegador y entra en `http://localhost` (o tu IP). Debería cargar la web desde la carpeta DAW.



3. Cambio de Puerto (80 al 2022)

Enunciado: Configurar Apache para que escuche en el puerto 2022 en lugar del estándar 80.

Configuración:

1. Editamos el archivo de puertos: `sudo nano /etc/apache2/ports.conf`

```
migueldaw@miguelillo:/var/www/html/DAW$ sudo nano /etc/apache2/ports.conf
```

- Cambia Listen 80 por Listen 2022.

```
migueldaw@miguelillo: /var/www/html/DAW
GNU nano 7.2 /etc/apache2/ports.conf *
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 2022

<IfModule ssl_module>
    Listen 443
</IfModule>

<IfModule mod_gnutls.c>
    Listen 443
</IfModule>
```

2. Editamos el virtual host: `sudo nano /etc/apache2/sites-available/000-default.conf`

```
migueldaw@miguelillo:/var/www/html/DAW$ sudo nano /etc/apache2/sites-available/000-default.conf
```

- Cambia `<VirtualHost *:80>` por `<VirtualHost *:2022>`.

```
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf *
<VirtualHost *:2022>
# The ServerName directive sets the request scheme, hostname and port to
# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
#ServerName www.example.com

ServerAdmin webmaster@localhost
DocumentRoot /var/www/html/DAW

# Available loglevels: trace8, ..., tracel, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn
```

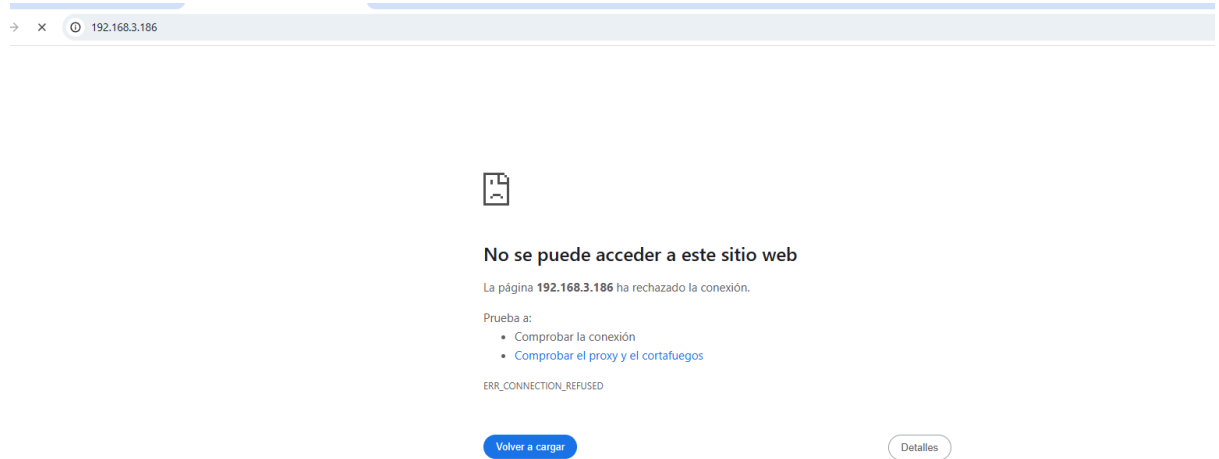
3. Reiniciamos: `sudo systemctl restart apache2`

```
migueldaw@miguelillo:/var/www/html/DAW$ sudo systemctl restart apache2
```

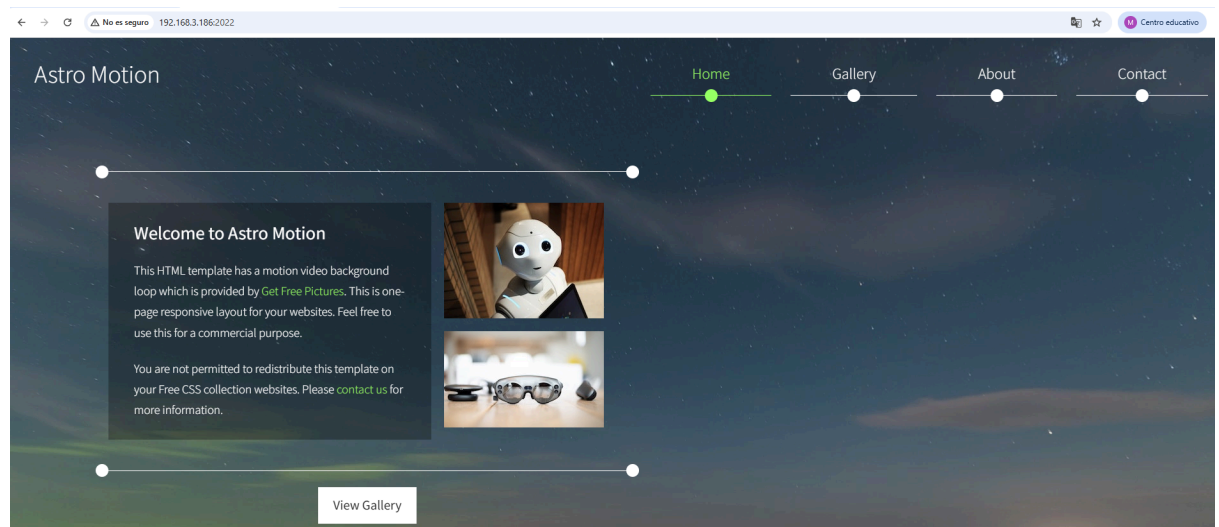
Prueba de funcionamiento:

- Intenta entrar por el puerto 80 (dará error) y luego por el 2022:
`http://localhost:2022`

puerto 80



puerto 2022



4. Prioridad de Ficheros (DirectoryIndex)

Enunciado: Renombrar `index.html` a `i.html`, observar el listado de directorio (error de seguridad), corregirlo mediante configuración y revertir el cambio.

Configuración y Pruebas:

1. Provocando el error:

- Ve a la carpeta: `cd /var/www/html/DAW`
- Renombra: `sudo mv index.html i.html`

```
migueldaw@miguelillo:/var/www/html/DAW$ sudo mv index.html i.html
```

- Recarga el navegador. Verás la lista de ficheros (tipo FTP).



2. Solucionando el error (Configuración):

- Edita el config: `sudo nano /etc/apache2/mods-enabled/dir.conf` (o en `apache2.conf`).

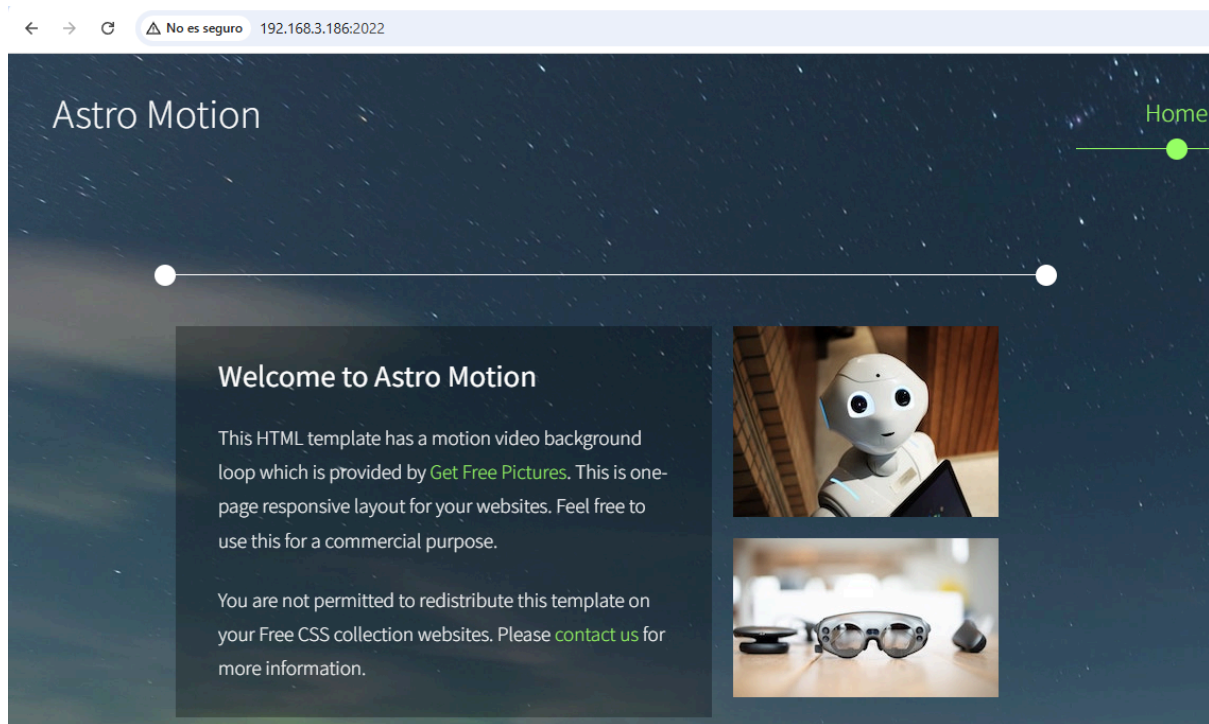
```
migueldaw@miguelillo:/var/www/html/DAW$ sudo nano /etc/apache2/mods-enabled/dir.conf
```

Añade `i.html` a la lista y vuelves a reiniciar apache:

```
DirectoryIndex index.html index.cgi index.pl index.php index.xhtml index.htm  
i.html
```

3. Comprobación:

- Recarga el navegador. Ahora debe verse la web, no la lista.



4. Revertir:

- Ejecuta: `sudo mv i.html index.html` para volver como estaba

5. Modificación del Fichero Hosts

Enunciado: Simular un dominio local `www.minombre.es` modificando el archivo `hosts`.

Configuración:

1. Edita el archivo `hosts`: `sudo nano /etc/hosts`


```
GNU nano 7.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 miguelillo

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters

[ Read 9 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Li
```

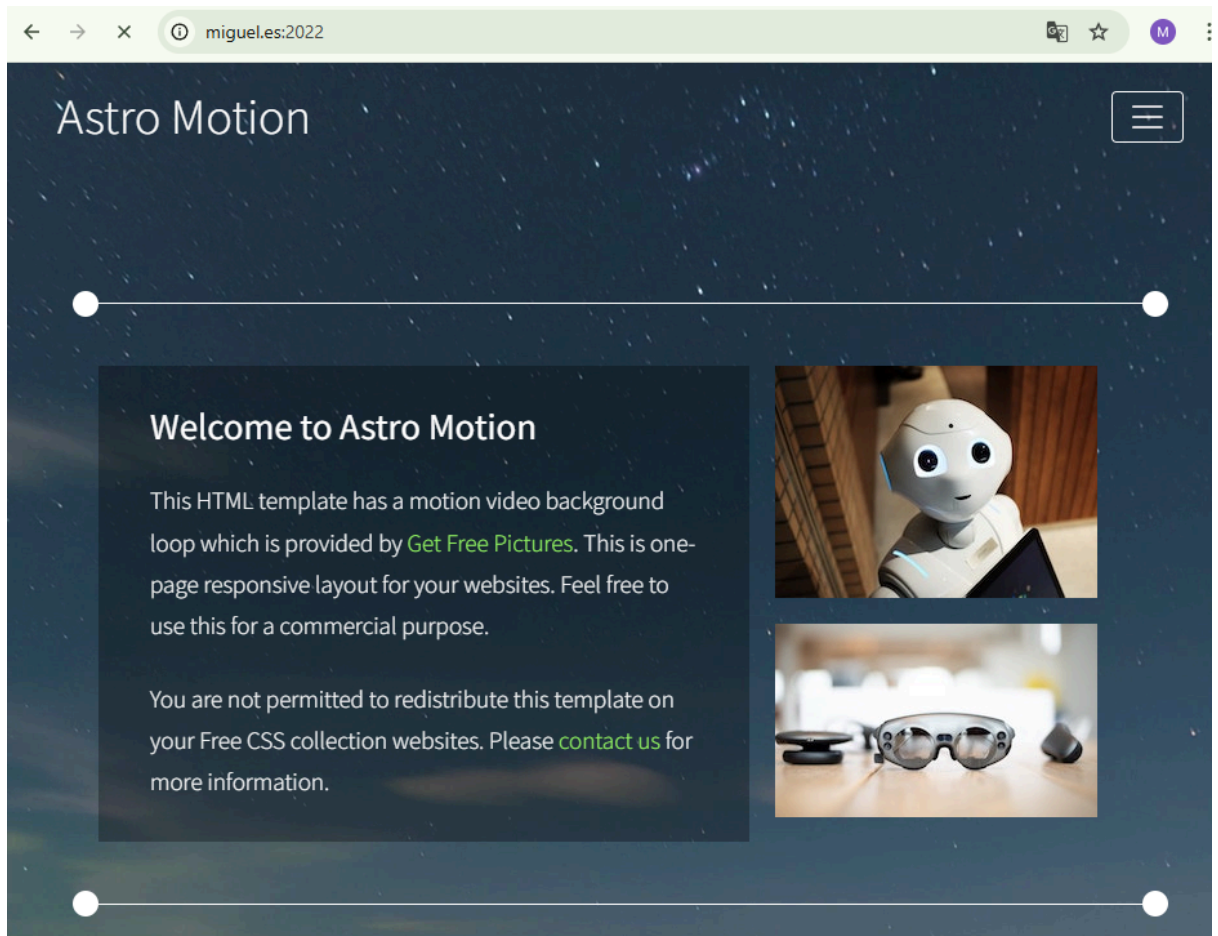
2. Añade la línea: 192.168.3.186 www.miguel.es

```
GNU nano 7.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 miguelillo
192.168.3.186 www.miguel.es

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
```

Prueba de funcionamiento:

1. Abre el navegador y ve a <http://www.miguel.es:2022>



6. Personalizar Error 404 y Ocultar Información

Enunciado: Cambiar la página de error 404 por defecto por una personalizada y ocultar la firma del servidor.

Configuración:

1. Ocultar info:
 - Edita: `sudo nano /etc/apache2/conf-available/security.conf` (o `apache2.conf` si no usas `conf-available`).

```
migueldaw@miguelillo:/var/www/html/DAW$ sudo nano /etc/apache2/conf-available/security.conf
```

- Cambia: `ServerTokens Prod` y `ServerSignature Off`.

```
# ServerTokens
# This directive configures
# Header. The default is 'Full'
# and compiled in modules.
# Set to one of: Full | Minimal | Off
# where Full conveys the most information.
#ServerTokens Minimal
ServerTokens Prod
#ServerTokens Full

#
# Optionally add a line containing
# name to server-generated
# listings, mod_status and
# documents or custom error
# Set to "EMail" to also include
# Set to one of: On | Off
#ServerSignature Off
ServerSignature Off
```

2. Página 404:

- Descarga una plantilla de error 404 y guárdala como error404.html en /var/www/html/DAW.
- Edita el sitio: `sudo nano /etc/apache2/sites-available/000-default.conf`

- `migueldaw@miguelillo:/var/www/html/DAW$ sudo nano /etc/apache2/sites-available/000-default.conf`
- Añade: `ErrorDocument 404 /error404.html`

```
GNU nano 7.2 /etc/apache2/sites-available/000-default.conf *
<VirtualHost *:2022>
    # The ServerName directive sets the request scheme, hostname and port that
    # the server uses to identify itself. This is used when creating
    # redirection URLs. In the context of virtual hosts, the ServerName
    # specifies what hostname must appear in the request's Host: header to
    # match this virtual host. For the default virtual host (this file) this
    # value is not decisive as it is used as a last resort host regardless.
    # However, you must set it for any further virtual host explicitly.
    #ServerName www.example.com

    ServerAdmin webmaster@localhost
    DocumentRoot /var/www/html/DAW

    # Available loglevels: trace8, ..., tracel, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined
    #ErrorDocument 404 /error404.html
    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

- Reinicia Apache.

Prueba de funcionamiento:

- Accede a una URL inventada: `http://www.minombre.es:2022/loquesea`



7. Bloquear Acceso por Navegador (User-Agent)

Enunciado: Denegar el acceso a un navegador específico (ej. Firefox o un móvil).

Configuración:

1. Activa `mod_rewrite` si no lo está: `sudo a2enmod rewrite` y reinicia.

```
</html>migueldaw@miguelillo:~$ sudo a2enmod
Your choices are: access_compat actions alias allowmethods asis auth_basic auth_digest auth_form authn_anon authn_cor
e authn_dbd authn_dbm authn_file authn_socache authnz_fcgi authnz_ldap authz_core authz_dbd authz_dbm authz_groupfile
authz_host authz_owner authz_user autoindex brotli buffer cache cache_disk cache_socache cern_meta cgi cgid charset_
lite data dav dav_fs dav_lock dbd deflate dialup dir dump_io echo env expires ext_filter file_cache filter_headers he
artbeat heartmonitor http2 ident imagemap include info lbmethod_bybusyness lbmethod_byrequests lbmethod_bytraffic lbm
ethod heartbeat ldap log_debug log_forensic lua macro md mime mime_magic mpm_event mpm_prefork mpm_worker negotiation
php8.3 proxy proxy_ajp proxy_balancer proxy_connect proxy_express proxy_fcgi proxy_fdpass proxy_ftp proxy_hcheck pro
xy_html proxy_http proxy_http2 proxy_scgi proxy_uwsgi proxy_wstunnel ratelimit reflector remoteip reqtimeout request
rewrite sed session session_cookie session_crypto session_dbd setenvif slotmem_plain slotmem_shm socache_dbm socache_
memcache socache_redis socache_shmcb spelling ssl status substitute suexec unique_id userdir usertrack vhost_alias xml
2enc
Which module(s) do you want to enable (wildcards ok)?
rewrite
Enabling module rewrite.
To activate the new configuration, you need to run:
    systemctl restart apache2
migueldaw@miguelillo:~$
```

2. Crea/Edita un archivo `.htaccess` en `/var/www/html/DAW/` o edita el `<Directory>` en `000-default.conf`.

```
sudo nano /etc/apache2/sites-available/000-default.conf
```

Añade:

Apache

RewriteEngine On

RewriteCond %{HTTP_USER_AGENT} Firefox [NC]

RewriteRule .* - [F,L]

```

# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
#ServerName www.example.com

ServerAdmin webmaster@localhost
DocumentRoot /var/www/html/DAW

# Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined
ErrorDocument 404 /error404.html

# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf

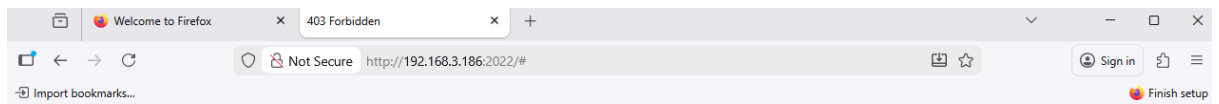
</VirtualHost>
Directory /var/www/html/DAW>
Options Indexes FollowSymLinks
AllowOverride All
Require all granted

# BLOQUEO DE FIREFOX
RewriteEngine On
RewriteCond %{HTTP_USER_AGENT} Firefox [NC]
RewriteRule .* - [F,L]
</Directory>

```

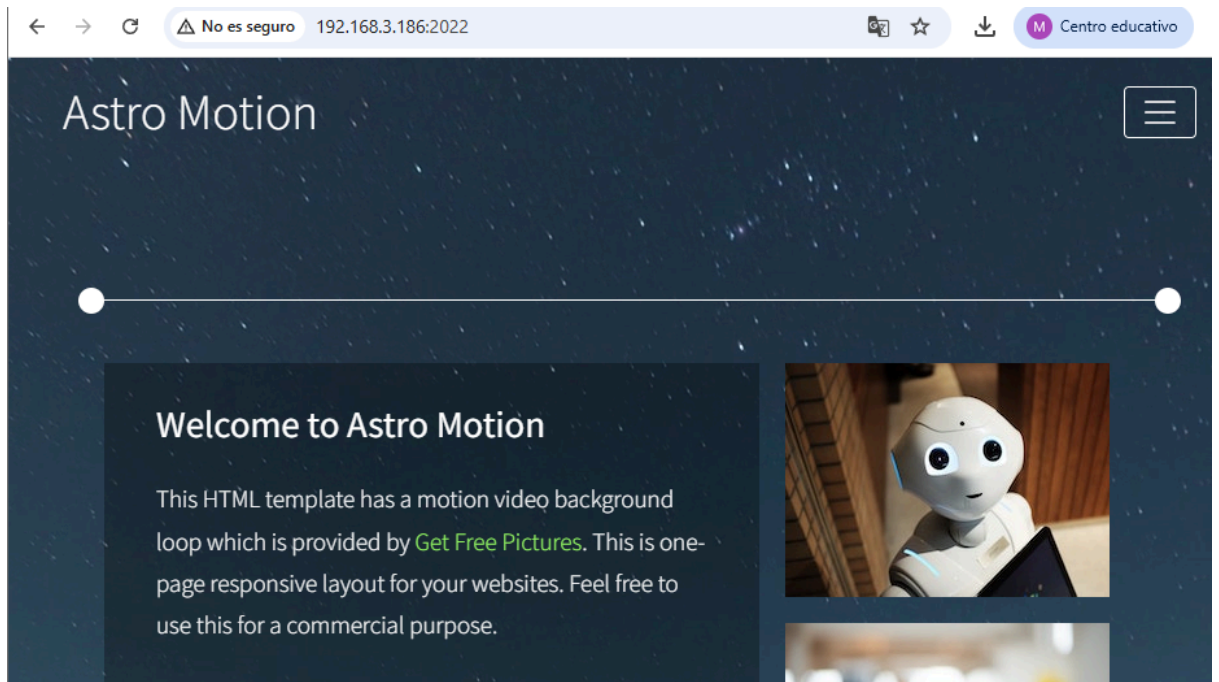
Prueba de funcionamiento:

- Intenta entrar con Firefox -> Error 403 Forbidden.
- Intenta entrar con Chrome/Edge -> Acceso correcto.



Forbidden

You don't have permission to access this resource.



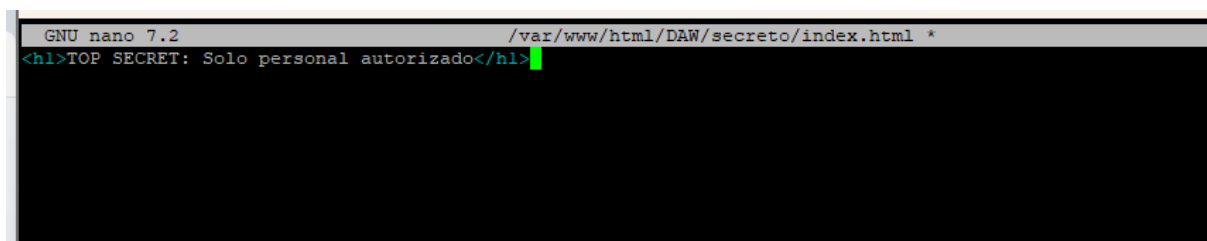
ANEXO EXTRA: Seguridad Avanzada

A. Bloquear Direcciones IP

Investigación: En Apache 2.4 se utiliza el módulo `mod_authz_core` mediante la directiva `Require`.

```
sudo mkdir /var/www/html/DAW/secreto
```

```
sudo nano /var/www/html/DAW/secreto/index.html
```



sudo htpasswd -c /etc/apache2/.htpasswd usuario1

contraseña usuario

```
migueldaw@miguelillo:~$ sudo mkdir /var/www/html/DAW/secreto
sudo nano /var/www/html/DAW/secreto/index.html
migueldaw@miguelillo:~$ sudo htpasswd -c /etc/apache2/.htpasswd usuario1
New password:
Re-type new password:
Adding password for user usuario1
migueldaw@miguelillo:~$
```

editamos el siguiente archivo

sudo nano /etc/apache2/sites-available/000-default.conf

```
# The ServerName directive sets the request scheme, hostname and port that
# the server uses to identify itself. This is used when creating
# redirection URLs. In the context of virtual hosts, the ServerName
# specifies what hostname must appear in the request's Host: header to
# match this virtual host. For the default virtual host (this file) this
# value is not decisive as it is used as a last resort host regardless.
# However, you must set it for any further virtual host explicitly.
#ServerName www.example.com

ServerAdmin webmaster@localhost
DocumentRoot /var/www/html/DAW

# Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn
<Directory /var/www/html/DAW>
Options Indexes FollowSymLinks
AllowOverride All
Require all granted

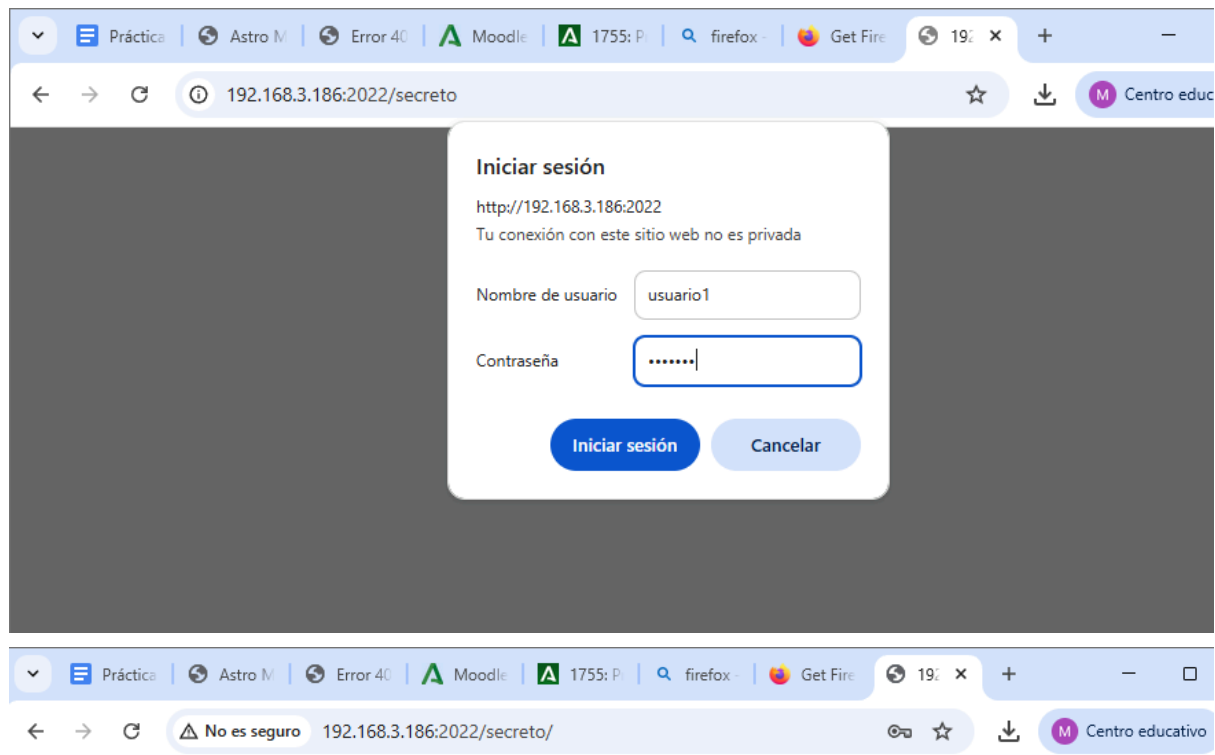
# BLOQUEO DE FIREFOX
RewriteEngine On
RewriteCond %{HTTP_USER_AGENT} Firefox [NC]
RewriteRule .* - [F,L]
</Directory>
<Directory "/var/www/html/DAW/secreto">
AuthType Basic
AuthName "Area Restringida"
AuthUserFile /etc/apache2/.htpasswd
Require valid-user
</Directory>

ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined
ErrorDocument 404 /error404.html
# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

y reiniciamos apache

```
Adding password for user usuariol
miguel daw@miguelillo:~$ sudo nano /etc/apache2/sites-available
miguel daw@miguelillo:~$ sudo nano /etc/apache2/sites-available
miguel daw@miguelillo:~$ sudo systemctl restart apache2
miguel daw@miguelillo:~$
```

nos vamos al apartado secreto en el navegador



TOP SECRET: Solo personal autorizado